ALI YOUSEFI

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Interactive portfolio: aliyousefi9403.github.io/3D-Room | YouTube: youtube.com/@Breachlab101

PROFILE

Final-year Computer Science student focused on AI security and applied AI. Lead project is AegisRAG, research on detecting knowledge base poisoning in RAG systems, written up as a preprint with Dr Hafizul Asad and Dr Nikos Komninos. Currently on the Hack The Box AI Red Teamer path alongside CompTIA Security+.

EXPERIENCE

AI Engineering Intern | Adaptive Swarm **2026 – Present**

- Develop and evaluate models on a node-based, self-adaptive AI platform, from data ingestion to signals scored on live markets, including a Bitcoin prediction model built from technical indicators.
- Give the founders structured feedback as an early engineering user, earning full access to the model and data layer.

Blockchain Security and Market Analyst | Batis Group **2025 – May 2026**

- Screened Ethereum and Tron tokens for rug-pull risk, checking ownership, mint rights, liquidity locks and holder spread.
- Owned company wallet and key management, redesigning storage and recovery to remove single points of failure.

Freelance Web Developer | International B2B Sales Representative, Kaveh Float **2022 – 2025**

- Delivered 2 business websites, including farazjam.com, from brief to launch as sole point of contact.
- Closed 4 international B2B deals across Italy, Germany, China and the Middle East, negotiating terms in a second language.

PROJECTS

Repositories private for security reasons unless linked. Walkthrough or live demo available on request.

AegisRAG | Detecting Knowledge Base Poisoning in RAG Systems **2026 – 2027**

Final Year Individual Project | Python, FastAPI, Qdrant, Next.js | Preprint: "Reading the Poison", with Dr Hafizul Asad and Dr Nikos Komninos

- Engineered a RAG security testbed over 200 MITRE ATT&CK and CAPEC passages and 44 benchmark questions; showed synonym-swap poisoning cuts correctness by 22% (0.714 to 0.555), mainly by displacing evidence.
- Created a relative-entropy detector reaching 0.95 AUROC that names the swapped word in 22 of 27 flagged passages.
- Halved poisoned text reaching the model (41% to 20%) with a filter at no measurable clean-accuracy cost; added an LTL runtime monitor with quarantine memory.
- Verified all 73 headline claims against 5,000 SHA-256 checksummed run records, backed by 285 passing tests.

The Room | Interactive 3D Portfolio with a Live Multi-Agent SOC **2026**

TypeScript, React Three Fiber, WebGL, WebSockets | aliyousefi9403.github.io/3D-Room

- Built a browser-based 3D room with 11 working stations, a page-flip paper reader and live BTC/ETH data over a Coinbase WebSocket.
- Designed Night Watch, a simulated SOC: 5 sliding-window detection rules mapped to MITRE ATT&CK, incident correlation, and 3 agents with A* navigation that triage and contain attacks launched by visitors.

CourtVision | Live Basketball Match Analyser

- Built a machine-learning system that tracks and re-identifies every player in a live game, recognises their actions, and generates post-game analysis per player plus a scouting plan against the next opponent.

AI Lease and Contract Analyser | Retrieval-Augmented Generation **2026**

- Constructing a RAG pipeline that maps each tenancy clause to its governing statute and flags non-standard, missing and unenforceable terms, scored against a hand-labelled test set. *Python, LLM APIs, embeddings.*

Aris | Voice and Gesture Controlled System Supervisor **2026**

- Created a sandboxed local assistant with a FastAPI WebSocket backend, a SQLite indexer over 12 git repos, a voice and gesture UI, and OAuth 2.0 with rotating refresh tokens. *Python, SQLite, MediaPipe.*

quantdesk | Systematic Trading Research Stack **2026**

- Engineered the research harness (DuckDB and Parquet store, cost model, backtester, walk-forward tests) and a multi-asset trend strategy with inverse-volatility sizing and volatility targeting. *Python.*

Pharmacy Ordering and Inventory System | Team Software Project (Project Manager) **2026**

- Led requirements, design and sprint planning, and co-wrote the C# front end and JWT-secured REST API.

SECURITY PRACTICE AND CERTIFICATION

In progress: Hack The Box AI Red Teamer path (primary focus), CompTIA Security+ and TryHackMe Red Teaming, due by January 2027.
Completed: TryHackMe Cyber Security 101 path plus web and cloud exploitation rooms.

- **AI red teaming:** practising prompt injection, data poisoning and output manipulation, feeding directly into AegisRAG.
- **Web:** found an exposed .git directory on a Flask host with gobuster and rebuilt the app source from loose objects.
- **Cloud:** abused an AWS Cognito pool issuing credentials to anonymous users to enumerate an over-permissive DynamoDB table.

TECHNICAL SKILLS

Security: LLM red teaming, RAG adversarial testing, detection engineering, web app testing, gobuster, Wireshark, git forensics, cloud IAM, Slither, OAuth 2.0.
Languages: Python, TypeScript, Java, C#, JavaScript, SQL, Bash.
Systems and Cloud: Linux (Kali, Ubuntu), macOS, Windows, TCP/IP, AWS (S3, Cognito, DynamoDB, IAM), Google Cloud, Azure, Docker.
Tools and AI: Git, FastAPI, React, Next.js, Three.js / React Three Fiber, Qdrant, REST APIs, WebSockets, MySQL, DuckDB, Parquet, LLM APIs, embeddings, vector search.

EDUCATION

City St George's, University of London | BSc (Hons) Computer Science **2024 – 2027**

Predicted 2:1 or above. Final year: Cloud Computing, Programming and Mathematics for AI, Theory of Computation, 45-credit Individual Project (AegisRAG). Earlier: Networks, Operating Systems, Algorithms.

ADDITIONAL

Content: write and code 2D animated cybersecurity explainers for the BreachLab YouTube channel. **Languages:** English and Turkish (fluent), Persian (native). **Right to work:** eligible for a placement year now, and for the UK Graduate visa (18 months, no sponsorship needed) after graduating in 2027. **Achievements:** Iran U18 National Basketball Team (#13); Iran U21 National Basketball Champion; UKMT Senior Mathematical Challenge Silver and Senior Kangaroo Certificate of Merit (2022).